

Is this a
Nut?



It looks like a **Nut**, and cracks like a **Nut**, so sure, but if you really want to know, ask C++.



No, but it shares some of the same **traits**. Don't worry, I'll protect you from using it improperly.



Not technically, but it implements the **Crackable** interface so you can treat it like one.



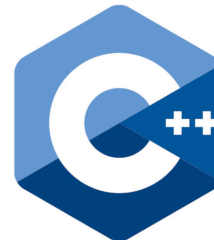
Doesn't matter. You can multiply it with **string** for all I care. It'll probably work.



I don't **classify** things. *You're* supposed to know what it is.



What Java said.



No, but we can work around that limitation if you're willing to risk the stability of the universe.

CS 252:

Advanced Programming Language Principles



Event-based Programming

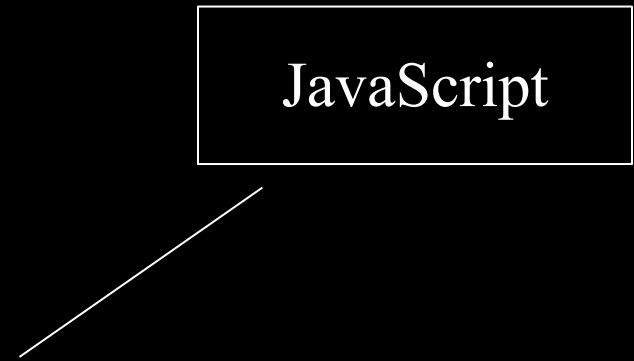
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Review JS Intro Lab

Inline JavaScript

```
<html>  
  <input  
    type='button'  
    onclick='alert("Hello!");'  
    value='Say hi' />  
</html>
```



A diagram consisting of a rectangular box labeled "JavaScript" with a line pointing from the box to the JavaScript code snippet within the HTML code block.

The "better" approach

```
<html>
  <input id='thebutton'
        type='button'
        value='Say hi' />
  <script type="text/javascript">
    var btn = document.
            getElementById('thebutton');
    btn.onclick = function() {
        alert('Groovy');
    };
  </script>
</html>
```

Note the
id attribute

Perhaps better still

```
<html>
  <input id='thebutton'
        type='button'
        value='Say hi' />
  <script type="text/javascript">
    var btn = document.
      getElementById('thebutton');
    function sayGroovy() {
      alert('Groovy');
    }
    btn.addEventListener('click', sayGroovy);
  </script>
</html>
```

```
function sayGroovy() {  
    alert('Groovy');  
}  
btn.addEventListener('click',  
                    sayGroovy);  
btn.addEventListener('click',  
    function() { alert("Bogus"); }  
);
```

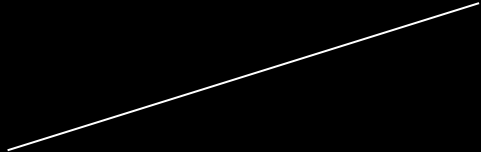
```
function sayGroovy() {  
    alert('Groovy');  
    btn.removeEventListener('click',  
                                sayGroovy);  
}  
  
btn.addEventListener('click',  
                    sayGroovy);  
  
btn.addEventListener('click',  
    function() { alert("Bogus"); }  
);
```


Note that JavaScript (in a browser) is single threaded.

An event runs to completion before the next event begins.

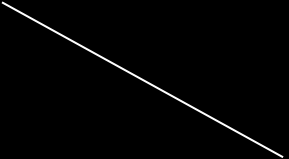
Importing events in Node.js

Choose whatever
name you like



```
var EE =  
    require('events').EventEmitter;
```

events is
a module



```
var EE = require('events').EventEmitter;
var ee = new EE();
```

```
die = false;
```

```
ee.on('die', function() {
  console.log("I'm melting!!! Oh, what a world...");
  die = true;
});
```

```
setTimeout(function() {
  ee.emit('die');
}, 100);
```

```
while (!die) {}
```

```
console.log('done');
```

TCP Server example

```
var net = require('net');
var eol = require('os').EOL;

var srvr = net.createServer();

srvr.on('connection', function(client) {
  client.write('Hello there!' + eol);
  client.end();
});

srvr.listen(9000);
```

```
$ node tcpserver.js
```

```
$
```

```
$ nc 127.0.0.1 9000
```

```
Trying 127.0.0.1...
```

```
Connected to localhost.
```

```
Escape character is '^]'.
```

```
Hello there!
```

```
Connection closed by foreign host.
```

```
$
```

Lab

- Lab: Write a chat server